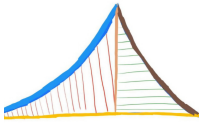
Math for EE



G. V. V. Sharma

About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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1. CLASS10 BOARDPAPER

A. Formulae

1.1.1. The *tangent* to a circle with centre $\mathbf{O}$ and radius r at the point $\mathbf{A}$ is perpendicular to the radius $\mathbf{OA}$,

$$(\mathbf{A}-\mathbf{O})^{\top}(\mathbf{P}-\mathbf{A})=0 \tag{1.1.1.1}$$

where $\mathbf{P}$ is any other point on the tangent.

1.1.2. The lengths of the two tangents from an external point $\mathbf{P}$ are equal,

$$\|\mathbf{P}-\mathbf{A}\|=\|\mathbf{P}-\mathbf{B}\| \tag{1.1.2.1}$$

1.1.3. The angle θ between $\mathbf{a}$ and $\mathbf{b}$ is given by

$$\cos\theta=\frac{\mathbf{a}^{\top}\mathbf{b}}{\|\mathbf{a}\|\,\|\mathbf{b}\|} \tag{1.1.3.1}$$

1.1.4. The sum of the angles of a quadrilateral is

$$360^{\circ} \tag{1.1.4.1}$$

1.1.5. The *mean* of x_i with frequencies f_i is

$$\bar{x}=\frac{\sum f_i x_i}{\sum f_i} \tag{1.1.5.1}$$

1.1.6. The *empirical relation* between the mean, median and mode is

$$3\,\text{median}=\text{mode}+2\,\text{mean} \tag{1.1.6.1}$$

1.1.7. The *discriminant* of $ax^2+bx+c=0$ is

$$D=b^2-4ac \tag{1.1.7.1}$$

The roots are equal if and only if $D=0$.

1.1.8. The *z-transform* of a pmf $p[n]$ is

$$H(z)=\sum_{n}p[n]z^{-n},\qquad H(1)=1 \tag{1.1.8.1}$$

1.1.9. If X and Y are independent, the pmf of $S=X+Y$ is the convolution of the pmfs, so

$$p_S[n]=p_X[n]*p_Y[n]\quad\overset{\mathcal{Z}}{\longleftrightarrow}\quad H_S(z)=H_X(z)H_Y(z) \tag{1.1.9.1}$$

and $\Pr(S=n)$ is the coefficient of z^{-n} in $H_S(z)$.

1.1.10. The *slant height* of a frustum of height h with radii R and r is

$$l=\sqrt{h^2+(R-r)^2} \tag{1.1.10.1}$$

1.1.11. A line that meets a circle at two points is called a *secant*. A line at distance d from the centre meets a circle of radius r in two points if $d<r$.

1.1.12. The tangents at the ends of a diameter are parallel. For $\mathbf{m}_1, \mathbf{m}_2$ the direction vectors of two lines,

$$\det\begin{pmatrix}\mathbf{m}_1 & \mathbf{m}_2\end{pmatrix} = 0 \tag{1.1.12.1}$$

1.1.13. The points $\mathbf{A}, \mathbf{B}, \mathbf{C}$ are collinear if

$$\det\begin{pmatrix}\mathbf{B}-\mathbf{A} & \mathbf{C}-\mathbf{A}\end{pmatrix} = 0 \tag{1.1.13.1}$$

1.1.14. If α, β are the zeroes of ax^2+bx+c, then

$$\alpha+\beta = -\frac{b}{a}, \qquad \alpha\beta = \frac{c}{a} \tag{1.1.14.1}$$

1.1.15. The volume of a cylinder of radius r and height h is

$$V = \pi r^2 h \tag{1.1.15.1}$$

1.1.16. If $\mathbf{O}$ is the midpoint of $\mathbf{A}$ and $\mathbf{B}$, then

$$\mathbf{B} = 2\mathbf{O} - \mathbf{A} \tag{1.1.16.1}$$

1.1.17. For positive integers p, q,

$$\text{LCM}\,(p,q) \times \text{HCF}\,(p,q) = pq \tag{1.1.17.1}$$

The LCM takes the highest power of each prime in p and q.

1.1.18. If $\triangle ABC \sim \triangle PQR$, the ratio of the areas is the square of the ratio of corresponding sides,

$$\frac{\text{ar}\,(ABC)}{\text{ar}\,(PQR)} = \left(\frac{BC}{QR}\right)^2 \tag{1.1.18.1}$$

1.1.19. In a right triangle, the *tangent* of an angle θ is

$$\tan\theta = \frac{\text{opposite}}{\text{adjacent}}, \qquad \tan 30^\circ = \frac{1}{\sqrt{3}} \tag{1.1.19.1}$$

1.1.20. *Euclid's division lemma*: for integers n and $d > 0$, there are unique q, r with

$$n = dq + r, \qquad 0 \le r < d, \qquad r = n \bmod d \tag{1.1.20.1}$$

1.1.21. The roots of $Ax^2+Bx+C=0$ are

$$x = \frac{-B \pm \sqrt{B^2-4AC}}{2A} \tag{1.1.21.1}$$

1.1.22. If $PQ \parallel BC$ in $\triangle ABC$, then $\triangle APQ \sim \triangle ABC$ and

$$\frac{AQ}{AC} = \frac{AP}{AB} = \frac{PQ}{BC} = k \tag{1.1.22.1}$$

The points are $\mathbf{P} = \mathbf{A} + k\,(\mathbf{B}-\mathbf{A})$ and $\mathbf{Q} = \mathbf{A} + k\,(\mathbf{C}-\mathbf{A})$.

1.1.23. The *probability* of an event E is

$$\Pr(E) = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}} \tag{1.1.23.1}$$

and $\Pr(E') = 1 - \Pr(E)$.

1.1.24. The *mode* of grouped data with modal class lower limit l, class size h, modal frequency f_1 and the frequencies f_0, f_2 of the classes before and after it, is

$$\text{mode} = l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2}\, h \tag{1.1.24.1}$$

1.1.25. The area of the triangle with vertices $\mathbf{A}, \mathbf{B}, \mathbf{C}$ is

$$\frac{1}{2}\left|\det\begin{pmatrix}\mathbf{B}-\mathbf{A} & \mathbf{C}-\mathbf{A}\end{pmatrix}\right| \tag{1.1.25.1}$$

1.1.26. Two linear equations in x, y can be written as $\mathbf{Mx} = \mathbf{c}$. If $\det \mathbf{M} \neq 0$,

$$\mathbf{x} = \mathbf{M}^{-1}\mathbf{c}, \qquad \begin{pmatrix} p & q \\ r & s \end{pmatrix}^{-1} = \frac{1}{ps - qr}\begin{pmatrix} s & -q \\ -r & p \end{pmatrix} \tag{1.1.26.1}$$

1.1.27. The tangent points $\mathbf{A}$ from an external point $\mathbf{P}$ to the circle with centre $\mathbf{O}$ and radius r satisfy

$$\|\mathbf{A} - \mathbf{O}\| = r, \qquad (\mathbf{A} - \mathbf{O})^{\top}(\mathbf{A} - \mathbf{P}) = 0 \tag{1.1.27.1}$$

so $\mathbf{A}$ lies on the circle with OP as diameter, and the tangent length is $\sqrt{OP^2 - r^2}$.

1.1.28. The area of a circle of diameter d and the volume of a coin of thickness t are

$$\text{area} = \pi\left(\frac{d}{2}\right)^2, \qquad \text{volume} = \text{area} \times t \tag{1.1.28.1}$$

1.1.29. *SSS and SAS similarity*: $\triangle ABC \sim \triangle PQR$ if

$$\frac{AB}{PQ} = \frac{BC}{QR} = \frac{CA}{RP} \quad \text{or} \quad \frac{AB}{PQ} = \frac{AC}{PR},\ \angle A = \angle P \tag{1.1.29.1}$$

Similar triangles have equal corresponding angles.

1.1.30. The diagonals of a parallelogram bisect each other. The median AD of $\triangle ABC$ joins $\mathbf{A}$ to the midpoint

$$\mathbf{D} = \frac{\mathbf{B} + \mathbf{C}}{2} \tag{1.1.30.1}$$

1.1.31. *Pythagoras theorem*: if $\angle A = 90^\circ$ in $\triangle ABC$, then

$$BC^2 = AB^2 + AC^2 \tag{1.1.31.1}$$

1.1.32. The *angle of depression* of an object seen from a height equals the angle of elevation of the top from the object. The values used are

$$\tan 45^\circ = 1, \qquad \tan 60^\circ = \sqrt{3} \tag{1.1.32.1}$$

1.1.33. The volumes of a cone and a hemisphere of radius r, and of a ring of height H with inner and outer radii r and R, are

$$V_{\text{cone}} = \frac{1}{3}\pi r^2 h, \quad V_{\text{hemisphere}} = \frac{2}{3}\pi r^3, \quad V_{\text{ring}} = \pi\left(R^2 - r^2\right)H \tag{1.1.33.1}$$

1.1.34. If A and B finish a work in T days together and alone in a and b days, the work

done in one day adds up,

$$\frac{1}{a}+\frac{1}{b}=\frac{1}{T} \tag{1.1.34.1}$$

1.1.35. *Division algorithm*: for polynomials f and $g \neq 0$,

$$f(x)=g(x)\,q(x)+r(x) \tag{1.1.35.1}$$

where $\deg r < \deg g$. If g divides f completely, $r=0$, and the zeroes of q are zeroes of f.

1.1.36. The *Newton–Raphson method* finds a zero of $f(x)$ from a starting point x_0 by

$$x_{n+1}=x_n-\frac{f(x_n)}{f'(x_n)} \tag{1.1.36.1}$$

1.1.37. The *median* of grouped data with $N=\sum f$, median class lower limit l, class size h, frequency f of the median class and cumulative frequency cf of the class before it, is

$$\text{median}=l+\frac{\frac{N}{2}-cf}{f}\,h \tag{1.1.37.1}$$

1.1.38. The *'less than' ogive* is drawn by plotting the points (upper class limit, cumulative frequency) and joining them. The median is the x-coordinate of the point of the ogive with $y=\frac{N}{2}$. Between two points $\mathbf{P}_1, \mathbf{P}_2$, a point of the ogive is

$$\mathbf{P}=\mathbf{P}_1+\lambda\left(\mathbf{P}_2-\mathbf{P}_1\right), \qquad 0\le\lambda\le 1 \tag{1.1.38.1}$$

B. Problems

1.2.5. In Figure-1, if tangents PA and PB from an external point $\mathbf{P}$ to a circle with centre $\mathbf{O}$, are inclined to each other at an angle of 80°, then $\angle AOB$ is equal to

a) 100°
b) 60°
c) 80°
d) 50°

(CBSE Class 10)

Solution: Since PA and PB are tangents, from (1.1.1.1),

$$(\mathbf{A}-\mathbf{O})^\top(\mathbf{P}-\mathbf{A})=0, \quad (\mathbf{B}-\mathbf{O})^\top(\mathbf{P}-\mathbf{B})=0 \tag{1.2.5.1}$$

so $\angle OAP=\angle OBP=90^\circ$. In the quadrilateral $OAPB$, from (1.1.4.1),

$$\angle AOB=360^\circ-90^\circ-90^\circ-80^\circ=100^\circ \tag{1.2.5.2}$$

This is verified with (1.1.3.1) in the code, which gives

$$\cos\theta=\frac{(\mathbf{A}-\mathbf{O})^\top(\mathbf{B}-\mathbf{O})}{\|\mathbf{A}-\mathbf{O}\|\,\|\mathbf{B}-\mathbf{O}\|}=\cos 100^\circ \tag{1.2.5.3}$$

Hence, the correct option is (a).

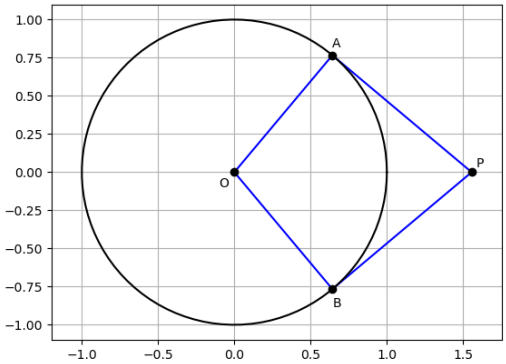


Fig. 1.2.5.1. Tangents PA and PB with $\angle APB = 80^\circ$

codes/q5.py

1.2.6. The mean and median of a distribution are 10 and 14 respectively. The value of mode is

a) 6
b) 22
c) 2
d) 20

(CBSE Class 10)

Solution: From (1.1.6.1),

$$\text{mode} = 3\,\text{median} - 2\,\text{mean} = 3\,(14) - 2\,(10) = 22 \tag{1.2.6.1}$$

Hence, the correct option is (b).

codes/q6.c

1.2.8. The value(s) of k for which the quadratic equation $3x^2 - kx + 3 = 0$ has equal roots, is (are)

a) 6
b) -6
c) ± 6
d) 9

OR

The discriminant of the quadratic equation $3\sqrt{3}x^2 + 10x + \sqrt{3} = 0$ is

a) ± 8
b) 8
c) $100 - 4\sqrt{3}$
d) 64

(CBSE Class 10)

Solution: Here $a = 3$, $b = -k$, $c = 3$. The roots are equal when $D = 0$, so from (1.1.7.1),

$$k^2 - 4\,(3)\,(3) = 0 \implies k^2 = 36 \implies k = \pm 6 \tag{1.2.8.1}$$

Hence, the correct option is (c).
OR Here $a = 3\sqrt{3}$, $b = 10$, $c = \sqrt{3}$. From (1.1.7.1),

$$D = 10^2 - 4\left(3\sqrt{3}\right)\left(\sqrt{3}\right) = 100 - 36 = 64 \tag{1.2.8.2}$$

Hence, the correct option is (d).

codes/q8.c

1.2.9. Two dice are thrown simultaneously. The probability that the sum of two numbers appearing on the top of the dice is less than 12, is

a) $\frac{1}{36}$
b) $\frac{35}{36}$
c) 0
d) 1

(CBSE Class 10)

Solution: Let X and Y be the numbers on the two dice and $S = X + Y$. The pmf of one die is $p[n] = \frac{1}{6}$ for $n = 1, \ldots, 6$. From (1.1.8.1),

$$H(z) = \frac{1}{6}\left(z^{-1} + z^{-2} + z^{-3} + z^{-4} + z^{-5} + z^{-6}\right) \tag{1.2.9.1}$$

Since the dice are independent, from (1.1.9.1),

$$H_S(z) = H(z)^2 = \frac{1}{36}\left(z^{-2} + 2z^{-3} + 3z^{-4} + \cdots + 6z^{-7} + \cdots + 2z^{-11} + z^{-12}\right) \tag{1.2.9.2}$$

The coefficient of z^{-12} is $\Pr(S = 12) = \frac{1}{36}$. Therefore

$$\Pr(S < 12) = 1 - \Pr(S = 12) = 1 - \frac{1}{36} = \frac{35}{36} \tag{1.2.9.3}$$

Hence, the correct option is (b).

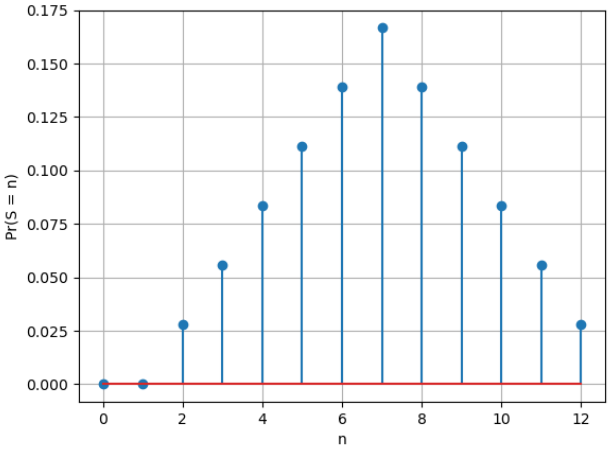


Fig. 1.2.9.1. Distribution of the sum of two dice

codes/q9.py

1.2.10. A frustum of a right circular cone which is of height 8 cm with radii of its circular ends as 10 cm and 4 cm, has its slant height equal to

a) 14 cm
b) 28 cm
c) 10 cm
d) $\sqrt{260}$ cm

(CBSE Class 10)

Solution: Here $h = 8$, $R = 10$, $r = 4$. From (1.1.10.1),

$$l = \sqrt{8^2 + (10-4)^2} = \sqrt{64+36} = \sqrt{100} = 10 \tag{1.2.10.1}$$

Hence, the correct option is (c).

codes/q10.c

1.2.13. A line intersecting a circle at two points is called a _____.

OR

The tangents drawn at the ends of a diameter of a circle are _____. (CBSE Class 10)

Solution: A line meeting a circle at two points is a *secant*. For the circle $x^2+y^2 = 1$ and the line $y = 0.5$, the distance $d = 0.5 < r = 1$, so the line meets the circle at two points.

OR Let $\mathbf{A} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$ be the ends of a diameter. The tangent directions

are perpendicular to $\mathbf{A}-\mathbf{O}$ and $\mathbf{B}-\mathbf{O}$,

$$\mathbf{m}_1 = \begin{pmatrix}0\\1\end{pmatrix}, \quad \mathbf{m}_2 = \begin{pmatrix}0\\-1\end{pmatrix} \tag{1.2.13.1}$$

From (1.1.12.1),

$$\det\begin{pmatrix}\mathbf{m}_1 & \mathbf{m}_2\end{pmatrix} = \begin{vmatrix}0 & 0\\1 & -1\end{vmatrix} = 0 \tag{1.2.13.2}$$

Hence, the answers are *secant* and *parallel*.

codes/q13.c

1.2.14. The value of m which makes the points $(0,0)$, $(2m,-4)$ and $(3,6)$ collinear, is _____. (CBSE Class 10)

Solution: Let $\mathbf{A} = \begin{pmatrix}0\\0\end{pmatrix}$, $\mathbf{B} = \begin{pmatrix}2m\\-4\end{pmatrix}$, $\mathbf{C} = \begin{pmatrix}3\\6\end{pmatrix}$. From (1.1.13.1),

$$\begin{vmatrix}2m & 3\\-4 & 6\end{vmatrix} = 12m + 12 = 0 \implies m = -1 \tag{1.2.14.1}$$

Hence, $m = -1$.

codes/q14.c

1.2.15. If α, β are zeroes of the polynomial $2x^2 - 5x - 4$, then $\frac{1}{\alpha} + \frac{1}{\beta} =$ _____. (CBSE Class 10)

Solution: From (1.1.14.1), $\alpha + \beta = \frac{5}{2}$ and $\alpha\beta = -2$. Then

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha+\beta}{\alpha\beta} = \frac{5/2}{-2} = -\frac{5}{4} \tag{1.2.15.1}$$

Hence, the answer is $-\frac{5}{4}$.

codes/q15.c

1.2.16. The capacity of a cylindrical glass tumbler is 125.6 cm^3. If the radius of the glass tumbler is 2 cm, then find its height. (Use $\pi = 3.14$) (CBSE Class 10)

Solution: From (1.1.15.1),

$$h = \frac{V}{\pi r^2} = \frac{125.6}{3.14 \times 4} = 10 \text{ cm} \tag{1.2.16.1}$$

Hence, the height is 10 cm.

codes/q16.c

1.2.17. A circle has its centre at $(4,4)$. If one end of a diameter is $(4,0)$, then find the coordinates of the other end. (CBSE Class 10)

Solution: Let $\mathbf{O} = \begin{pmatrix}4\\4\end{pmatrix}$ and $\mathbf{A} = \begin{pmatrix}4\\0\end{pmatrix}$. The centre is the midpoint of the diameter, so from (1.1.16.1),

$$\mathbf{B} = 2\begin{pmatrix}4\\4\end{pmatrix} - \begin{pmatrix}4\\0\end{pmatrix} = \begin{pmatrix}4\\8\end{pmatrix} \tag{1.2.17.1}$$

Hence, the other end is $(4,8)$.

codes/q17.c

1.2.18. If two positive integers p and q can be expressed as $p = ab^3$ and $q = a^2b$; a and b being prime numbers, then find LCM of (p, q). (CBSE Class 10)
Solution: The LCM takes the highest power of each prime from (1.1.17.1),

$$\text{LCM}\,(p, q) = a^2b^3 \tag{1.2.18.1}$$

For $a = 2$, $b = 3$: $p = 54$, $q = 12$, HCF $= 6$ and

$$\text{LCM} = \frac{54 \times 12}{6} = 108 = 2^2 \times 3^3 \tag{1.2.18.2}$$

Hence, LCM $(p, q) = a^2b^3$.

codes/q18.c

1.2.19. It is given that $\triangle ABC \sim \triangle PQR$ with $\frac{BC}{QR} = \frac{1}{3}$, then find the value of $\frac{\text{ar}(PRQ)}{\text{ar}(ACB)}$. (CBSE Class 10)
Solution: From (1.1.18.1),

$$\frac{\text{ar}\,(ABC)}{\text{ar}\,(PQR)} = \left(\frac{1}{3}\right)^2 = \frac{1}{9} \tag{1.2.19.1}$$

Since ar (PRQ) = ar (PQR) and ar (ACB) = ar (ABC),

$$\frac{\text{ar}\,(PRQ)}{\text{ar}\,(ACB)} = 9 \tag{1.2.19.2}$$

Hence, the answer is 9.

codes/q19.c

1.2.20. In Figure-2, a tower stands vertically on the ground. From a point on the ground, which is 80 m away from the foot of the tower, the angle of elevation of the tower is found to be $30°$. Find the height of the tower. (CBSE Class 10)
Solution: Let $\mathbf{B}$ be the foot of the tower, $\mathbf{A}$ the point on the ground and $\mathbf{C}$ the top,

$$\mathbf{B} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} 80 \\ 0 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} 0 \\ h \end{pmatrix} \tag{1.2.20.1}$$

From (1.1.19.1),

$$\tan 30° = \frac{BC}{AB} = \frac{h}{80} \implies h = \frac{80}{\sqrt{3}} = \frac{80\sqrt{3}}{3} \approx 46.19 \text{ m} \tag{1.2.20.2}$$

Hence, the height of the tower is $\frac{80\sqrt{3}}{3}$ m.

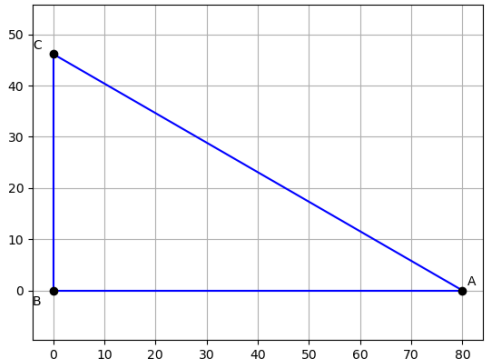


Fig. 1.2.20.1. Tower BC with angle of elevation 30° at A

codes/q20.py

1.2.21. Show that any positive odd integer is of the form $4q+1$ or $4q+3$ for some integer q. (CBSE Class 10)

Solution: Let n be a positive integer. From (1.1.20.1) with $d = 4$,

$$n = 4q + r, \qquad r \in \{0, 1, 2, 3\} \tag{1.2.21.1}$$

For $r = 0$ and $r = 2$, $n = 4q$ and $n = 4q + 2$ are even. So if n is odd, $r = 1$ or $r = 3$,

$$n = 4q + 1 \quad \text{or} \quad n = 4q + 3 \tag{1.2.21.2}$$

The code checks that $n \bmod 4 \in \{1, 3\}$ for every odd $n < 100$.

codes/q21.c

1.2.22. In Figure-3, $PQ \parallel BC$, $PQ = 3$ cm, $BC = 9$ cm and $AC = 7.5$ cm. Find the length of AQ. (CBSE Class 10)

Solution: Since $PQ \parallel BC$, from (1.1.22.1),

$$k = \frac{PQ}{BC} = \frac{3}{9} = \frac{1}{3} \tag{1.2.22.1}$$

Then

$$AQ = k\,AC = \frac{1}{3}\,(7.5) = 2.5 \text{ cm} \tag{1.2.22.2}$$

In the code, $\mathbf{Q} = \mathbf{A} + k\,(\mathbf{C} - \mathbf{A})$ gives $\|\mathbf{Q} - \mathbf{A}\| = 2.5$. Hence, $AQ = 2.5$ cm.

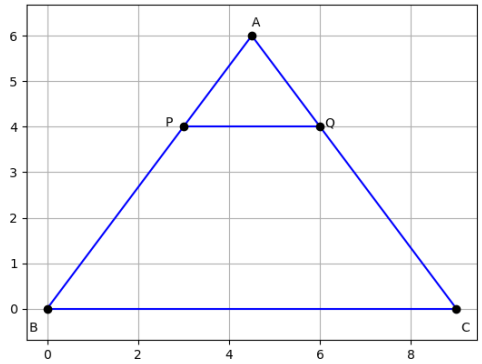


Fig. 1.2.22.1. $PQ \parallel BC$ in $\triangle ABC$

codes/q22.py

1.2.23. Solve $9x^2 - 6a^2x + a^4 - b^4 = 0$ using quadratic formula. (CBSE Class 10)

Solution: Here $A = 9$, $B = -6a^2$, $C = a^4 - b^4$. The discriminant is

$$B^2 - 4AC = 36a^4 - 36\left(a^4 - b^4\right) = 36b^4 \tag{1.2.23.1}$$

From (1.1.21.1),

$$x = \frac{6a^2 \pm 6b^2}{18} = \frac{a^2 \pm b^2}{3} \tag{1.2.23.2}$$

For $a = 2$, $b = 1$ the roots are $\frac{5}{3}$ and 1, as the code shows. Hence, $x = \frac{a^2+b^2}{3}$ or $x = \frac{a^2-b^2}{3}$.

codes/q23.c

1.2.24. A jar contains 18 marbles. Some are red and others are yellow. If a marble is drawn at random from the jar, the probability that it is red is $\frac{2}{3}$. Find the number of yellow marbles in the jar.

OR

A die is thrown twice. What is the probability that

a) 5 will come up at least once, and

b) 5 will not come up either time?

(CBSE Class 10)

Solution: From (1.1.23.1), the number of red marbles is

$$18 \times \frac{2}{3} = 12 \tag{1.2.24.1}$$

so the number of yellow marbles is $18 - 12 = 6$.

OR The die has $6 \times 6 = 36$ outcomes. A 5 does not come up either time in $5 \times 5 = 25$

outcomes, so from (1.1.23.1),

$$\Pr\left(\text{no } 5\right) = \frac{25}{36}, \qquad \Pr\left(\text{at least one } 5\right) = 1 - \frac{25}{36} = \frac{11}{36} \tag{1.2.24.2}$$

Hence, the answers are 6 yellow marbles, and $\frac{11}{36}$ and $\frac{25}{36}$.

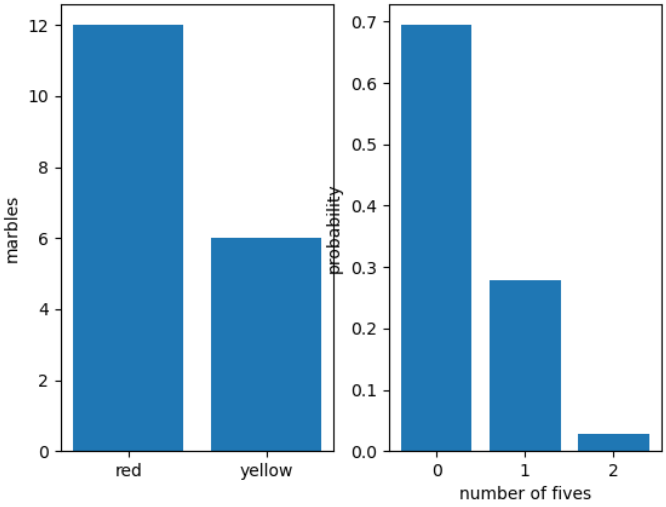


Fig. 1.2.24.1. Marbles in the jar and the number of fives in two throws

codes/q24.py

1.2.26. Find mode of the following distribution:

Class	25–30	30–35	35–40	40–45	45–50	50–55
Frequency	20	36	53	40	28	14

(CBSE Class 10)

Solution: The highest frequency is 53, so the modal class is 35–40. Here $l = 35$, $h = 5$, $f_1 = 53$, $f_0 = 36$, $f_2 = 40$. From (1.1.24.1),

$$\text{mode} = 35 + \frac{53 - 36}{2\left(53\right) - 36 - 40} \times 5 = 35 + \frac{17}{30} \times 5 = 35 + \frac{17}{6} = \frac{227}{6} \approx 37.83 \tag{1.2.26.1}$$

Hence, the mode is 37.83.

codes/q26.c

1.2.28. If the points $\mathbf{A}\left(2, 0\right)$, $\mathbf{B}\left(6, 1\right)$ and $\mathbf{C}\left(p, q\right)$ form a triangle of area 12 sq. units (positive only) and $2p + q = 10$, then find the values of p and q. (CBSE Class 10)

Solution: The vertices are

$$\mathbf{A} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 6 \\ 1 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} p \\ q \end{pmatrix} \tag{1.2.28.1}$$

From (1.1.25.1), taking the area as $+12$,

$$\frac{1}{2}\begin{vmatrix} 4 & p-2 \\ 1 & q \end{vmatrix} = \frac{1}{2}(4q - p + 2) = 12 \implies -p + 4q = 22 \tag{1.2.28.2}$$

Together with $2p + q = 10$, the system in matrix form is

$$\begin{pmatrix} -1 & 4 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} 22 \\ 10 \end{pmatrix} \tag{1.2.28.3}$$

The determinant of the matrix is $-1 - 8 = -9$, so

$$\begin{pmatrix} p \\ q \end{pmatrix} = \frac{1}{-9}\begin{pmatrix} 1 & -4 \\ -2 & -1 \end{pmatrix} \begin{pmatrix} 22 \\ 10 \end{pmatrix} = \frac{1}{-9}\begin{pmatrix} -18 \\ -54 \end{pmatrix} = \begin{pmatrix} 2 \\ 6 \end{pmatrix} \tag{1.2.28.4}$$

Hence, $p = 2$ and $q = 6$.

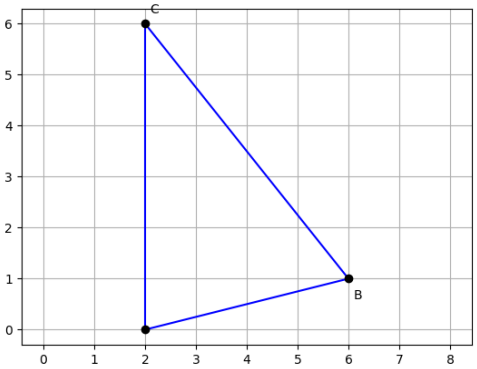


Fig. 1.2.28.1. Triangle ABC with $\mathbf{C} = (2, 6)$

codes/q28.py

1.2.30. Solve the following equations for x and y using cross-multiplication method:

$$(ax - by) + (a + 4b) = 0, \qquad (bx + ay) + (b - 4a) = 0 \tag{1.2.30.1}$$

(CBSE Class 10)

Solution: In matrix form,

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -(a + 4b) \\ 4a - b \end{pmatrix} \tag{1.2.30.2}$$

The determinant is $a^2+b^2 \neq 0$. From (1.1.26.1),

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{a^2+b^2}\begin{pmatrix} a & b \\ -b & a \end{pmatrix}\begin{pmatrix} -a-4b \\ 4a-b \end{pmatrix} = \frac{1}{a^2+b^2}\begin{pmatrix} -a^2-b^2 \\ 4a^2+4b^2 \end{pmatrix} = \begin{pmatrix} -1 \\ 4 \end{pmatrix} \tag{1.2.30.3}$$

Hence, $x=-1$ and $y=4$.

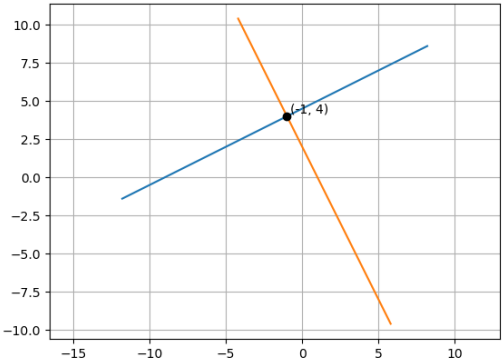


Fig. 1.2.30.1. The two lines meet at $(-1,4)$ (plotted for $a=1$, $b=2$)

codes/q30.py

1.2.31. Draw a circle of radius 2.5 cm. Take a point P outside the circle at a distance of 7 cm from the centre. Then construct a pair of tangents to the circle from point P. (CBSE Class 10)

Solution: Take $\mathbf{O} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, $\mathbf{P} = \begin{pmatrix} 7 \\ 0 \end{pmatrix}$ and $r = 2.5$. From (1.1.27.1), a tangent point $\mathbf{A} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ satisfies

$$\mathbf{A}^\top\mathbf{A} = r^2, \qquad \mathbf{A}^\top(\mathbf{A}-\mathbf{P}) = 0 \implies \mathbf{A}^\top\mathbf{P} = r^2 \tag{1.2.31.1}$$

So $7x_1 = 6.25$ and

$$x_1 = \frac{25}{28}, \qquad x_2 = \pm\sqrt{6.25 - x_1^2} \approx \pm 2.34 \tag{1.2.31.2}$$

The tangent points are $\mathbf{A} = \begin{pmatrix} 0.89 \\ 2.34 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 0.89 \\ -2.34 \end{pmatrix}$. The tangent length is

$$\|\mathbf{P}-\mathbf{A}\| = \sqrt{7^2 - 2.5^2} = \sqrt{42.75} \approx 6.54 \text{ cm} \tag{1.2.31.3}$$

Joining $\mathbf{P}$ to $\mathbf{A}$ and $\mathbf{B}$ gives the pair of tangents.

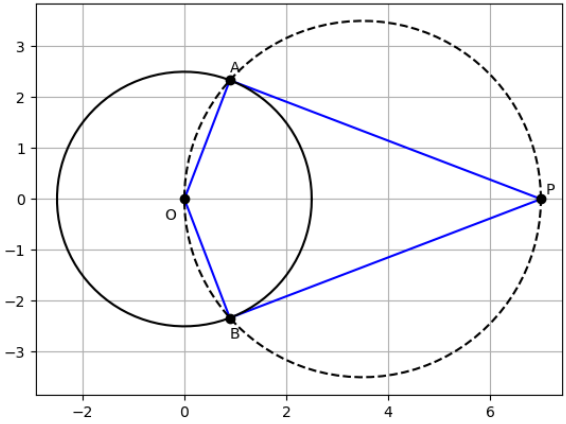


Fig. 1.2.31.1. Tangents PA and PB, with the circle on OP as diameter dashed

codes/q31.py

1.2.32. A mint moulds four types of copper coins A, B, C and D whose diameters vary from 0.5 cm to 5 cm. The first coin A has a diameter of 0.7 cm. The second coin B has double the diameter of coin A and from then onwards the diameters increase by 50%. Thickness of each coin is 0.25 cm.

a) Fill in the diameters of the coins required in the table.

b) Complete the table of area of one face and volume. [Use $\pi = \frac{22}{7}$]

(CBSE Class 10)

Solution: (i) The diameter of B is $2 \times 0.7 = 1.4$ cm. Then C is $1.5 \times 1.4 = 2.1$ cm and D is $1.5 \times 2.1 = 3.15$ cm.

Type of Coin	Diameter (in cm)
A	0.7
B	1.4

(ii) From (1.1.28.1), for coin B,

$$\text{area} = \frac{22}{7}(0.7)^2 = 1.54 \text{ cm}^2, \qquad \text{volume} = 1.54 \times 0.25 = 0.385 \text{ cm}^3 \quad (1.2.32.1)$$

Type of Coin	Area (in cm^2) of one face	Volume (in cm^3)
A	0.385	0.09625
B	1.54	0.385

codes/q32.c

1.2.35. Sides AB and AC and median AD of $\triangle ABC$ are respectively proportional to sides PQ and PR and median PM of $\triangle PQR$. Show that $\triangle ABC \sim \triangle PQR$.

OR

In Figure-4, BN and CM are medians of a $\triangle ABC$ right-angled at A. Prove that $4\left(BN^2 + CM^2\right) = 5BC^2$. (CBSE Class 10)

Solution: Given:

$$\frac{AB}{PQ} = \frac{AC}{PR} = \frac{AD}{PM} \tag{1.2.35.1}$$

Construction: Extend AD to E such that $DE = AD$ and join BE, CE. Extend PM to F such that $MF = PM$ and join QF, RF.

Proof: In the quadrilateral $ABEC$, D is the midpoint of both BC and AE (from (1.1.30.1)), so the diagonals bisect each other and $ABEC$ is a parallelogram. Hence

$$BE = AC, \qquad CE = AB \tag{1.2.35.2}$$

Similarly $PQFR$ is a parallelogram, so $QF = PR$ and $RF = PQ$. From the given condition,

$$\frac{AB}{PQ} = \frac{BE}{QF} = \frac{AE}{PF} \tag{1.2.35.3}$$

because $\frac{BE}{QF} = \frac{AC}{PR}$ and $\frac{AE}{PF} = \frac{2AD}{2PM} = \frac{AD}{PM}$. By SSS from (1.1.29.1),

$$\triangle ABE \sim \triangle PQF \implies \angle BAE = \angle QPF \tag{1.2.35.4}$$

In the same way,

$$\frac{AC}{PR} = \frac{CE}{RF} = \frac{AE}{PF} \implies \triangle ACE \sim \triangle PRF \implies \angle CAE = \angle RPF \tag{1.2.35.5}$$

Adding the two angle equalities,

$$\angle BAC = \angle QPR \tag{1.2.35.6}$$

Now $\frac{AB}{PQ} = \frac{AC}{PR}$ and $\angle A = \angle P$, so by SAS from (1.1.29.1),

$$\triangle ABC \sim \triangle PQR \tag{1.2.35.7}$$

The code checks the three ratios $\frac{AB}{PQ}$, $\frac{AC}{PR}$, $\frac{AD}{PM}$ for a sample pair of triangles.

OR Take $\mathbf{A} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} x \\ 0 \end{pmatrix}$, $\mathbf{C} = \begin{pmatrix} 0 \\ y \end{pmatrix}$. From (1.1.30.1) the midpoints are

$$\mathbf{M} = \begin{pmatrix} x/2 \\ 0 \end{pmatrix}, \qquad \mathbf{N} = \begin{pmatrix} 0 \\ y/2 \end{pmatrix} \tag{1.2.35.8}$$

Then

$$BN^2 = \|\mathbf{N} - \mathbf{B}\|^2 = x^2 + \frac{y^2}{4} \tag{1.2.35.9}$$

$$CM^2 = \|\mathbf{M} - \mathbf{C}\|^2 = \frac{x^2}{4} + y^2 \tag{1.2.35.10}$$

Adding and using (1.1.31.1),

$$BN^2 + CM^2 = \frac{5}{4}\left(x^2 + y^2\right) = \frac{5}{4}BC^2 \implies 4\left(BN^2 + CM^2\right) = 5BC^2 \tag{1.2.35.11}$$

For $x = 8$, $y = 4$, the code gives $4\,(68 + 32) = 400 = 5 \times 80$. Hence proved.

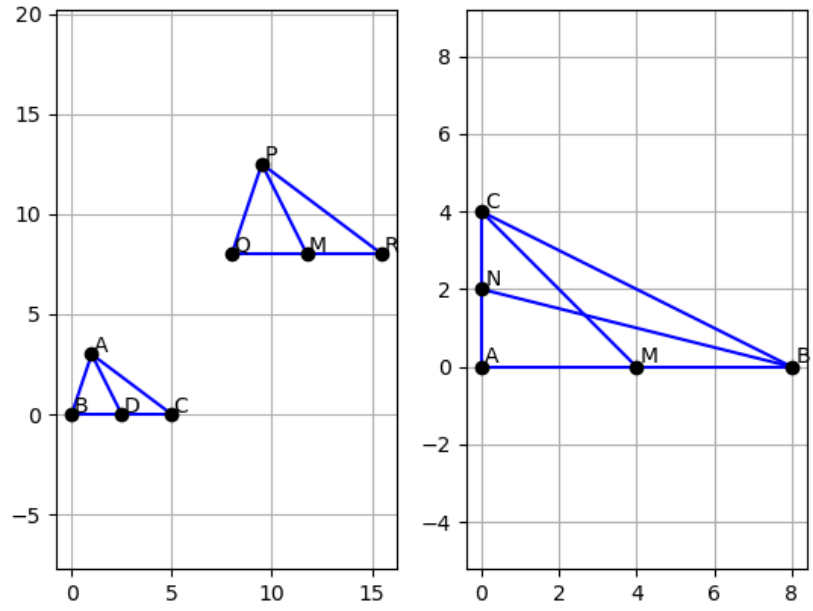


Fig. 1.2.35.1. Left: $\triangle ABC$ and $\triangle PQR$ with medians. Right: Figure-4

codes/q35.py

1.2.36. The angles of depression of the top and bottom of a tower as seen from the top of a $60\sqrt{3}$ m high cliff are 45° and 60° respectively. Find the height of the tower. (Use $\sqrt{3} = 1.73$) (CBSE Class 10)

Solution: Let $H = 60\sqrt{3}$ be the height of the cliff, d the distance between the cliff and the tower, and h the height of the tower. From (1.1.32.1), for the bottom of the tower,

$$\tan 60^\circ = \frac{H}{d} \implies d = \frac{60\sqrt{3}}{\sqrt{3}} = 60 \text{ m} \tag{1.2.36.1}$$

For the top of the tower,

$$\tan 45^\circ = \frac{H-h}{d} = 1 \implies h = H - d = 60\sqrt{3} - 60 = 60\left(\sqrt{3} - 1\right) \tag{1.2.36.2}$$

With $\sqrt{3} = 1.73$,

$$h = 60 \times 0.73 = 43.8 \text{ m} \tag{1.2.36.3}$$

Hence, the height of the tower is 43.8 m.

codes/q36.c

1.2.37. A well of diameter 3 m is dug 14 m deep. The earth taken out of it has been spread evenly all around it in the shape of a circular ring of width 4 m to form a platform. Find the height of the platform. (Take $\pi = \frac{22}{7}$)

OR

In Figure-5, a solid toy is in the form of a hemisphere surmounted by a right circular cone. The height of the cone is 2 cm and the diameter of the base is 4 cm. Determine the volume of the toy. (Take $\pi = 3.14$) (CBSE Class 10)

Solution: The well is a cylinder with $r = 1.5$ m and depth 14 m. From (1.1.15.1),

$$V = \frac{22}{7}(1.5)^2(14) = 99 \text{ m}^3 \tag{1.2.37.1}$$

The ring has inner radius 1.5 m and outer radius $1.5 + 4 = 5.5$ m. From (1.1.33.1),

$$\frac{22}{7}\left(5.5^2 - 1.5^2\right)H = 99 \implies \frac{22}{7}(28)H = 99 \implies H = \frac{99}{88} = 1.125 \text{ m} \tag{1.2.37.2}$$

Hence, the height of the platform is 1.125 m.

OR The cone and the hemisphere have $r = 2$ cm and the cone has $h = 2$ cm. From (1.1.33.1),

$$V = \frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3 = 3.14\left(\frac{8}{3} + \frac{16}{3}\right) = 3.14 \times 8 = 25.12 \text{ cm}^3 \tag{1.2.37.3}$$

Hence, the volume of the toy is 25.12 cm^3.

codes/q37.c

1.2.38. A and B jointly finish a piece of work in 15 days. When they work separately, A takes 16 days less than the number of days taken by B to finish the same piece of work. Find the number of days taken by B to finish the work. (CBSE Class 10)

Solution: Let B take x days, so A takes $x - 16$ days. From (1.1.34.1),

$$\frac{1}{x} + \frac{1}{x-16} = \frac{1}{15} \implies 15(2x - 16) = x(x - 16) \implies x^2 - 46x + 240 = 0 \tag{1.2.38.1}$$

From (1.1.21.1),

$$x = \frac{46 \pm \sqrt{2116 - 960}}{2} = \frac{46 \pm 34}{2} = 40 \text{ or } 6 \tag{1.2.38.2}$$

For $x = 6$, A would take $6 - 16 < 0$ days, which is not possible. Hence, B takes 40 days.

codes/q38.c

1.2.39. If the polynomial $f(x) = 3x^4 - 9x^3 + x^2 + 15x + k$ is completely divisible by $3x^2 - 5$, then find the value of k. Using the quotient, so obtained, find two zeroes of the polynomial.

OR

Find all the zeroes of the polynomial $x^4 - 8x^3 + 23x^2 - 28x + 12$ if two of its zeroes are 2 and 3. (CBSE Class 10)

Solution: Dividing $f(x)$ by $3x^2-5$ from (1.1.35.1),

$$3x^4-9x^3+x^2+15x+k=\left(3x^2-5\right)\left(x^2-3x+2\right)+(k+10) \quad (1.2.39.1)$$

Since the division is complete, the remainder is zero,

$$k+10=0 \implies k=-10 \quad (1.2.39.2)$$

The quotient is $x^2-3x+2=(x-1)(x-2)$, so two zeroes of $f(x)$ are 1 and 2. With $f(x)=x^2-3x+2$, (1.1.36.1) reads

$$x_{n+1}=x_n-\frac{x_n^2-3x_n+2}{2x_n-3} \quad (1.2.39.3)$$

For $x_0=0$, $x_1=\frac{2}{3}$ and the iterates converge to 1. For $x_0=3$, $x_1=\frac{7}{3}$ and the iterates converge to 2, as the code shows. Hence, $k=-10$ and the zeroes are 1 and 2.

OR Since 2 and 3 are zeroes, $(x-2)(x-3)=x^2-5x+6$ divides the polynomial. From (1.1.35.1),

$$x^4-8x^3+23x^2-28x+12=\left(x^2-5x+6\right)\left(x^2-3x+2\right) \quad (1.2.39.4)$$

The quotient $x^2-3x+2=(x-1)(x-2)$ gives the other zeroes. Newton–Raphson from $x_0=0$, 1.5 and 4 converges to 1, 2 and 3 respectively. Hence, all the zeroes are 1, 2, 2, 3.

codes/q39.py

1.2.40. Draw 'less than' ogive for the following distribution and hence find its median.

Class	Frequency
20–30	10
30–40	8
40–50	12
50–60	24
60–70	6
70–80	25
80–90	15

(CBSE Class 10)

Solution: The points of the 'less than' ogive are the upper class limits with their cumulative frequencies,

Upper limit	30	40	50	60	70	80	90
cf	10	18	30	54	60	85	100

Plotting these points and joining them gives the ogive. Here $N=100$, so $\frac{N}{2}=50$. The point of the ogive with $y=50$ lies between

$$\mathbf{P}_1=\begin{pmatrix}50\\30\end{pmatrix}, \qquad \mathbf{P}_2=\begin{pmatrix}60\\54\end{pmatrix} \quad (1.2.40.1)$$

From (1.1.38.1), the y-coordinate gives

$$30 + \lambda\,(54 - 30) = 50 \implies \lambda = \frac{20}{24} = \frac{5}{6} \tag{1.2.40.2}$$

and the x-coordinate, which is the median, is

$$x = 50 + \frac{5}{6}\,(60 - 50) = 50 + \frac{25}{3} = \frac{175}{3} \approx 58.33 \tag{1.2.40.3}$$

This agrees with (1.1.37.1) for the median class 50–60, where $l = 50$, $cf = 30$, $f = 24$, $h = 10$,

$$\text{median} = 50 + \frac{50 - 30}{24} \times 10 \approx 58.33 \tag{1.2.40.4}$$

Hence, the median is 58.33.

codes/q40.c